

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY (EGLE)

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: September 30, 2025

SUBJECT: Cadillac Wastewater Treatment Plant (WWTP)
Local Limits Review
Submission Number: HQF-TQCJ-GZWEB
National Pollutant Discharge Elimination System (NPDES) Permit Number MI0020257

Pursuant to a request dated September 23, 2025, we are providing theoretical water quality-based effluent limitations (WQBELs) for the Cadillac WWTP facility. The facility is currently authorized to continuously discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3984. A design flow of 3.2 million gallons per day (MGD) was used in determining final effluent limitations and monitoring requirements in the current NPDES permit, but this flow is not to be considered a limitation or actual capacity.

The Clam River has a lowest 95 percent exceedance flow, harmonic mean flow, and 90dQ10 low flow of 2.6, 13, and 3.4 cubic feet per second (cfs), respectively (EGLE Low Flow Database File Number 10194 [Wexford County]). The Clam River is designated as warmwater where Outfall 001 discharges into and is not protected as a public drinking water supply source.

A total hardness value of 70 milligrams per liter (mg/l) was used for this review. This total hardness value was collected by EGLE staff upstream of Outfall 001 on August 19, 2021 at Latitude 44.263376, Longitude -85.4011 (STORET Number 830054).

The theoretical water quality-based effluent limits (WQBELs) for the parameters you requested are provided below in micrograms per liter (ug/l) and pounds per day (lbs./day).

Parameter	CAS Number	Daily Maximum WQBEL (ug/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (ug/l)	Monthly Average Load (lbs./day)
Beryllium	7440417	66	1.8	4.1	0.11
Benzidine #	92875	49	1.3	0.12	0.0032
Bromomethane	74839	75	2.0	4.8	0.13
Phenol	108952	6,800	180	510	14

= carcinogen.

The Preliminary Effluent Limitations (PEL) file is saved in the MiEnviro Portal under the following filename: IPP Local Limits PEL_MI0020257_9-30-2025.xls

Please contact me at (517) 290-6424 if you have any questions regarding the recommendations above.